

Homework 3

Introduction to Functions

Name: _____

Instructions

Use a **pen**. Write questions and surprises in the margin in a different color. Some statements in this homework are intentionally imprecise, and some contradict what you already know — find them and flag them in the margin. If a topic confuses you, you can use the internet or AI to help. AI can be a good teacher if you ask it specific questions. **DO NOT** use AI to solve the problems without trying yourself first.

Remember to write comments in the margin and answer the entire question. You are a mathematician, your answers should be detailed and convincing to anyone that reads it. **Do not leave ambiguity** (remember what ambiguous means? if not, search it up).

Part 1 — Foundation

1. Let $f(x) = 3x + 4$. Fill in the table, then answer the questions.

x	$f(x)$
0	
1	
-1	
2	
$\frac{1}{3}$	

- (a) What is $f(f(1))$?

Answer: _____

- (b) Find x such that $f(x) = 10$.

Answer: _____

(c) Find x such that $f(x) = 0$.

Answer: _____

2. For each rule, decide whether it defines a valid function from A to B . If it does, state whether it is injective and/or surjective. If it does not, explain why not.

(a) $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4, 5\}$: $1 \mapsto 2$, $2 \mapsto 4$, $3 \mapsto 5$.

(b) $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4, 5\}$: $1 \mapsto 3$, $2 \mapsto 3$, $3 \mapsto 3$.

(c) $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4, 5\}$: $1 \mapsto 2$, $2 \mapsto 4$, $2 \mapsto 5$.

(d) $A = \{1, 2, 3\}$, $B = \{1, 2, 3\}$: each number maps to one more than itself, $x \mapsto x + 1$.

(e) $A = \{1, 2, 3\}$, $B = \{a, b, c\}$: $1 \mapsto a$, $2 \mapsto b$.

3. Domain, codomain, and range.

(a) $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = x^2 + 2$. State the domain, codomain, and range.

(b) $h: \{1, 2, 3\} \rightarrow \{3, 6, 11\}$ defined by $h(x) = x^2 + 2$. State the domain, codomain, and range. Is h surjective? You used the same formula as (a) — explain why the answers differ.

(c) *No formula required.* A Minecraft furnace smelts each input into one output:

$\text{smelt}: \{\text{iron ore, gold ore, sand, raw beef}\} \longrightarrow \{\text{iron ingot, gold ingot, glass, steak}\},$

$\text{iron ore} \mapsto \text{iron ingot}, \text{ gold ore} \mapsto \text{gold ingot}, \text{ sand} \mapsto \text{glass}, \text{ raw beef} \mapsto \text{steak}.$

State the domain, codomain, and range of smelt. Can you write smelt as a formula $\text{smelt}(x) = \dots$? What does your answer say about the claim “every function is a formula”?

4. Fill in the blanks, then answer the True/False items (justify each in one sentence).

(a) The range of a function is always a _____ of its codomain.

(b) In symbols, for $f: A \rightarrow B$, the range $R(f)$ satisfies $R(f) \subseteq$ _____.

(c) A function is surjective exactly when its range equals its _____.

T / F For every function, the range is a subset of the codomain.

T / F For every function, the range equals the codomain.

T / F Every function is surjective onto its own range (shrink the codomain to the range and it becomes surjective).

(d) Give a function $f: \{1, 2\} \rightarrow \{a, b, c\}$ whose range is a **proper** subset of its codomain. Write it as ordered pairs, and state the range.

5. For finite sets, the sizes $|A|$ and $|B|$ alone determine what kinds of functions can exist. Answer True/False and justify each in one sentence.

T / F If $|A| > |B|$, then no function $f: A \rightarrow B$ can be injective.

T / F It is possible for $f: A \rightarrow B$ to be injective when $|A| \geq |B|$.

T / F If $|A| < |B|$, then no function $f: A \rightarrow B$ can be surjective.

T / F If $|A| = |B|$ (finite) and f is injective, then f must also be surjective.

T / F If $f: A \rightarrow B$ is bijective, then $|A| = |B|$.

(a) Complete the table. For each pair of sizes, circle whether each type of function from A to B is **possible**.

Sizes	Injective possible?	Surjective possible?	Bijective possible?
$ A = 2, B = 4$	Yes No	Yes No	Yes No
$ A = 4, B = 2$	Yes No	Yes No	Yes No
$ A = 3, B = 3$	Yes No	Yes No	Yes No

6. One Function, Four Outfits. A function can be dressed four ways — an **arrow diagram**, a set of **ordered pairs**, a **table**, and a **graph**. Here is one function, given as ordered pairs:

$$f: \{-1, 0, 1, 2\} \rightarrow \mathbb{Z}, \quad f = \{(-1, 1), (0, -1), (1, -3), (2, -5)\}.$$

(a) Complete the table.

x	-1	0	1	2
$f(x)$				

(b) Draw the arrow diagram, and plot the points on a grid.

Arrow diagram	Graph

(c) Find the **rule**. $f(x) =$ _____

(d) Is f injective? Is f surjective onto $\mathbb{Z}$? For each, name the representation that made it easiest to see, and say why.

7. Is It a Function? Reading the Graph. The **vertical line test**: a graph in the xy -plane defines y as a function of x exactly when no vertical line meets it more than once. For each, decide whether it defines y as a function of x — that is, whether there is a function f with $y = f(x)$. Circle Yes / No and justify in one sentence.

(a) The points $\{(1, 2), (2, 4), (1, 5), (3, 4)\}$. Yes No

(b) The horizontal line $y = 3$. Yes No

(c) The vertical line $x = 3$. Yes No

(d) The parabola $y = x^2 - 1$. Yes No

(e) The sideways parabola $x = y^2$. Yes No

(f) The circle $x^2 + y^2 = 16$. Yes No

(g) In your own words, why does the vertical line test work? (What does a vertical line hitting the graph twice say about a single input?)

Reading — The Two Arrows: $\rightarrow$ and $\mapsto$

Function statements use two different arrows, and they are **not** interchangeable (do you know what interchangeable means?).

The long arrow $\rightarrow$ joins two *sets*. Writing $f: A \rightarrow B$ says “ f is a function from the set A to the set B .” It names the domain and the codomain. Read it “ f , from A to B .”

The bar arrow $\mapsto$ joins two *elements*. Writing $1 \mapsto a$ says “the input 1 is sent to the output a ” — the very same statement as $f(1) = a$. Read it “1 maps to a .” The little bar at the tail reminds you it starts at one specific element.

So one fact can be written three equivalent ways:

$$f(1) = a \quad \Longleftrightarrow \quad 1 \mapsto a \quad \Longleftrightarrow \quad (1, a) \in f.$$

Putting it together, a line like

$$p: \{1, 2, 3\} \rightarrow \{a, b, c, d\}, \quad 1 \mapsto a, \quad 2 \mapsto b, \quad 3 \mapsto c$$

first declares the domain and codomain (the $\rightarrow$ part), then lists where each input goes (the $\mapsto$ part). It is the same function as the ordered-pair set $\{(1, a), (2, b), (3, c)\}$.

8. Injective, Surjective, Bijective — Spot Them, Then Build Them.

Part 1 — Label each function as **injective**, **surjective**, **bijective**, or **none**.

(a) $p: \{1, 2, 3\} \rightarrow \{a, b, c, d\}: 1 \mapsto a, 2 \mapsto b, 3 \mapsto c$.

(b) $q: \{1, 2, 3\} \rightarrow \{a, b\}: 1 \mapsto a, 2 \mapsto b, 3 \mapsto a$.

(c) $r: \{1, 2, 3\} \rightarrow \{a, b, c\}: 1 \mapsto c, 2 \mapsto a, 3 \mapsto b$.

(d) $s: \{1, 2, 3\} \rightarrow \{a, b, c\}: 1 \mapsto a, 2 \mapsto a, 3 \mapsto b$.

Part 2 — Build your own (write each as a set of ordered pairs).

(e) An **injective but not surjective** function $\{1, 2\} \rightarrow \{a, b, c\}$.

(f) A **surjective but not injective** function $\{1, 2, 3\} \rightarrow \{a, b\}$.

(g) A **bijective** function $\{1, 2, 3\} \rightarrow \{x, y, z\}$.

(h) ★ Try to build a **bijective** function $\{1, 2, 3\} \rightarrow \{a, b\}$. Explain exactly what goes wrong.

Part 2 — Application

1. Three Bedrooms. Back to the Minecraft build. You are now carpeting **three** bedrooms. Every room is 10 blocks long. Bedroom 1 is 6 blocks wide and Bedroom 2 is 8 blocks wide — both finished. Bedroom 3 has an undecided width w . Each floor block needs one carpet block, and each carpet block takes 2 wool.

(a) Write a function $f(w)$ giving the **total** wool needed for all three rooms.

(b) Evaluate $f(5)$. What does this number mean in context?

(c) You have collected exactly 480 wool and want to use all of it. Find w such that $f(w) = 480$. What does this answer mean in context?

(d) State the natural domain of f in the Minecraft context. Is f surjective onto the positive integers $\mathbb{Z}^+$? Explain, and give one positive integer that is never an output.

2. Speaking Functions.

(a) Read $f(x) = 2x^2 - 5$ aloud as one complete English sentence. Write what you would say.

(b) A classmate says: “ $f(x)$ means f times x .” Are they right? Correct them in one sentence.

(c) Read $f(g(x))$ aloud as a complete English sentence. What do you think this expression means?

3. Functions Live Inside $A \times B$. Last week you learned the Cartesian product $A \times B$. A function $f: A \rightarrow B$ is a special subset of $A \times B$ in which each element of A appears as a first coordinate exactly once.

Let $A = \{1, 2\}$ and $B = \{p, q, r\}$.

(a) List every element of $A \times B$.

(b) How many ordered pairs are in $A \times B$? (Recall $|A \times B| = |A| \cdot |B|$.)

(c) For each subset, circle whether it is a function from A to B . For each that is not, say why.

(i) $\{(1, p), (2, r)\}$ function? Yes No

(ii) $\{(1, p), (1, q)\}$ function? Yes No

(iii) $\{(1, p)\}$ function? Yes No

(d) ★ How many functions are there from $A = \{1, 2\}$ to $B = \{p, q, r\}$ in total? Explain your count — do not just guess.

Looking Ahead — Why Functions, and Where We Go Next

Read this. Some of it is meant to feel unfinished — that is the point. It also contains at least one statement that is not quite right; find it and flag it in the margin. Where you get stuck, write the question down and we will resolve it next class.

Here is the story so far. We started with **sets** (collections of objects), built **ordered pairs** and the **Cartesian product** $A \times B$, and discovered that a **function** is just a special subset of $A \times B$. Then we learned to measure functions with **domain**, **codomain**, and **range**, and to classify them as **injective**, **surjective**, or **bijective**. Next we ask three new questions: how do functions *combine*, how do we *undo* them, and how can they be used to *measure infinity*?

1. Combining Functions — Composition

Think of two machines in a line. A furnace turns **ore** into an **ingot**; a crafting table turns an **ingot** into a **tool**. Feed ore in one end and a tool comes out the other — you have chained two functions. If f is “smelt” and g is “craft,” the combined machine is the **composition** $g \circ f$, read “ g of f ,” where $(g \circ f)(x) = g(f(x))$.

Try it. Let $f(x) = x + 1$ and $g(x) = 2x$. Compute $(g \circ f)(3)$, then $(f \circ g)(3)$. Did you get the same answer both ways?

2. Undoing Functions — Inverses

In the Three Bedrooms problem you ran the wool function *backwards*: you knew the output (480 wool) and recovered the input (w). That “backwards machine” is called the **inverse**, written f^{-1} . An inverse takes each output back to the input it came from. Because every function can be undone, every function has an inverse.

Try it. For $f(w) = 20w + 280$, describe in words the steps of the backwards machine that turns a wool count back into a width.

Answer this. What if two different inputs gave the *same* output? Could the backwards machine still know which input to return to? Explain.

Reading — Proofs and Why a Picture Is Not Enough

*Read this before Part 3. The Challenge problems ask you to **prove** things. So first, what is a proof, and why do mathematicians insist on them instead of just drawing a picture?*

A claim about a function is a claim about *every* input

When we say “ f is injective” we are promising something about **every** pair of inputs: no matter which two different inputs you pick, their outputs differ. When we say “ f is surjective” we promise that **every** element of the codomain gets hit. These are “for all” statements — they sweep over the whole domain or codomain, which may be infinite.

An arrow diagram, a table, or a sketched graph shows you only the cases you actually drew. For a function on a tiny finite set like $\{1, 2, 3\}$ you can draw **all** the arrows, and then the picture really does check every case — that is an honest proof by exhaustion, because you looked at everything. But the moment the domain is large or infinite — $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{R}$ — you can never draw all the arrows or plot the whole graph. The picture becomes a **sample**, and a sample cannot settle a claim about infinitely many inputs.

Pictures can fool you

A picture is where **ideas** come from — it shows you what to believe and where to hunt for trouble. A **proof** is what makes you **certain**: a chain of logical steps, built from **definitions**, that forces the conclusion for every case at once.

The engine: reason about an *arbitrary* element

Here is the trick that lets a few lines of writing cover infinitely many cases. Instead of testing specific inputs, we say “let a and b be **arbitrary**” — we assume nothing about which elements they are. Whatever we then prove about a and b must hold for **all** of them, because we never used anything special about them. One argument, every case. This is the whole reason a proof can beat infinitely many pictures.

A note on the $\square$. The little box at the end of a proof is a full stop for arguments: it announces “the proof is finished. Everything we promised has now been shown.” It is shorthand for **QED**, the Latin *quod erat demonstrandum*, meaning “which was to be demonstrated.” Mathematicians once spelled out the letters Q.E.D.; today most use the hollow box $\square$ (or a filled-in $\blacksquare$), a habit popularized by the mathematician Paul Halmos, who borrowed it from magazine typography. The symbol carries no mathematical content — it is purely a sign to the reader, marking exactly where the reasoning ends so that nothing after it gets mistaken for part of the argument.

Proving injectivity

Definition. f is injective if $f(a) = f(b)$ forces $a = b$ (recall this is the contrapositive of the definition of injectivity).

Blueprint. We approach this by first supposing $f(a) = f(b)$ for arbitrary a, b in the domain. Then we manipulate using the rule of f . Then if we can conclude $a = b$, we have proved f is injective.

Worked example. Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 3x + 4$.

Claim: f is injective.

Proof. Suppose $f(a) = f(b)$. Then $3a + 4 = 3b + 4$, so $3a = 3b$, therefore $a = b$. Since a, b were arbitrary, f is injective. $\square$

Notice what no picture could do: this single argument covered every one of the infinitely many pairs of real numbers.

Disproving injectivity takes only ONE counterexample. To show $g(x) = x^2$ is **not** injective we need no general argument — just one clash: $g(2) = 4 = g(-2)$, yet $2 \neq -2$. A single counterexample destroys a “for all” claim.

Proving surjectivity

Definition. $f: A \rightarrow B$ is surjective if for every $b \in B$ there exists some $a \in A$ with $f(a) = b$.

Template. Take an *arbitrary* target $y \in B$. *Produce* an input $x \in A$ (usually by solving $f(x) = y$) and check that $f(x) = y$.

Worked example. Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 3x + 4$.

Claim: f is surjective.

Proof. Let $y \in \mathbb{R}$ be arbitrary. Set $x = \frac{y-4}{3}$, which is a real number, so it lies in the domain. Then

$$f(x) = 3 \cdot \frac{y-4}{3} + 4 = (y-4) + 4 = y.$$

So every target y has a corresponding element in the domain. Therefore f is surjective. $\square$

You may want to re-read the example above several times and convince yourself that it is true (ask yourself questions like “why did we need to show $f(x) = y$?”)

Surjectivity depends on the codomain. Suppose we defined $g: \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = x^2$. Then g is **not** surjective because the target $y = -1$ has no real corresponding input x with $x^2 = -1$. So -1 is never hit (one unreachable target is enough to disprove “onto”). If we shrunk the codomain to $[0, \infty)$ then our function g becomes surjective. A surjectivity proof is always a statement about the **declared** codomain.

Proving bijectivity

Definition. f is bijective if it is both injective and surjective.

So a bijectivity proof is simply **two** proofs stacked: prove injective, then prove surjective. Our $f(x) = 3x + 4$ is both, so it is a bijection — and that is exactly the condition under which f has an inverse (the “undo” machine from the preview).

Part 3 — Challenge

1. Proving the Wool Function. In Part 2 you built the three-bedroom wool function $f(w) = 20w + 280$. Take both the domain and codomain to be the positive integers $\mathbb{Z}^+$ (whole blocks of width, whole counts of wool).

(a) Is f injective? Prove it.

(b) Is f surjective? Prove or disprove.

(c) Is f bijective? Prove or disprove.

(d) In one sentence, say what injectivity means **in the Minecraft context**: what does it guarantee about the relationship between a width and a wool count?

2. Odd One Out. Consider $f: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(n) = 2n + 1$.

(a) Is f injective? Prove it.

(b) Is f surjective? Prove or disprove. Remember that finding a counterexample counts as a disproof.

(c) Describe the range of f in words. If you kept the same rule but shrank the codomain to make f surjective, what set would the new codomain be?

3. The Finite Trap. Try to build a function $f: \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4\}$ that is injective but **not** surjective. Explain what happened and why.

4. The Collatz Function. Define $C: \mathbb{N} \rightarrow \mathbb{N}$ by

$$C(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ 3n + 1 & \text{if } n \text{ is odd.} \end{cases}$$

Starting from any positive integer, apply C over and over.

(a) Compute the sequence starting at $n = 7$. Stop when you reach 1. You can use a calculator.

(b) Compute the first 10 terms starting at $n = 27$. (Just list them; you do not need to reach 1.)

(c) Is C injective? Either prove it or find a counterexample. (Hint: can an even number and an odd number ever land on the same output?)

(d) Write a **conjecture** about what always seems to happen when you apply C repeatedly, starting from any positive integer.

5. ★ As Many Evens as Whole Numbers? Let $E = \{2, 4, 6, 8, \dots\}$ be the set of positive even numbers. Define the function $f: \mathbb{N} \rightarrow E$ by $f(n) = 2n$.

(a) Is f injective? Prove it.

(b) Is f surjective onto E ? Prove it.

(c) We will say that a bijection between two sets means they have the same size. But E is only *part* of $\mathbb{N}$. What strange thing does this say about the size of $\mathbb{N}$ compared to E ?

(d) ★ For **finite** sets, a proper subset always has strictly fewer elements — so if $B \subset A$ you would expect no way to match A one-to-one into B . In parts (a)–(b) you matched $\mathbb{N}$ one-to-one with E , the even numbers, even though E is a proper subset of $\mathbb{N}$ (it leaves out every odd number). What does this mapping show about the rule “a proper subset is strictly smaller” for infinite sets? What does it suggest about judging the size of an infinite set?

Part 4 — Notebook Artifact

1. If you used the internet or AI for help, describe how you used it:
